

Multiple Choice (10 marks)

1. Cellular uptake of cholesterol occurs through internalization of low density lipoprotein-cholesterol complexes. Cholesterol uptake involves all of the following EXCEPT
 - (a) cell-surface receptors
 - (b) adhesion plaques
 - (c) endosome formation
 - (d) receptor-mediated endocytosis
2. The target of digestion of a nucleosome dimer to nucleosome monomers by DNase is
 - (a) histones H 2 A, H 2 B, H 3, and H 4
 - (b) linker DNA
 - (c) histone H 2 A and H 2 B only
 - (d) the nucleosome linker
 - (e) histone H 3 only
3. What are the different types of point mutations?
 - (a) Frame shift mutations and nucleotide repeats
 - (b) Base substitutions and small insertions and deletions
 - (c) Base additions and large insertions
 - (d) Silent mutations and missense mutations
 - (e) Duplication mutations and triplet repeat expansions
4. What factors affect the rate of transpiration?
 - (a) Temperature, humidity, and air currents
 - (b) Plant species, root depth, and wind speed
 - (c) Soil fertility, irrigation methods, and cloud cover
 - (d) Water salinity, leaf color, and time of day
 - (e) Leaf surface area, stem length, and moon phase
5. In what way does Lamarck's theory of acquired characteristics not agree with present evidence?
 - (a) Acquired characteristics are inherited through a blending of parental traits
 - (b) Acquired characteristics are inherited but do not influence evolution
 - (c) Acquired characteristics are inherited randomly, regardless of their use
 - (d) Acquired characteristics cannot be inherited
 - (e) Acquired characteristics are inherited based on use and disuse

True / False (10 marks)

Answer True or False

6. True or False: The quantity of DNA in a eukaryotic cell typically doubles during the S phase of the cell cycle.
7. True or False: Proprioceptors are sensory receptors that provide information about body position and movement, allowing coordination without visual input.
8. True or False: A cell is a complex network of fibers that support body structures, functioning primarily as a support system.
9. True or False: The glomerular filtration rate is primarily influenced by arterial blood pressure and the diameter of the afferent arterioles.
10. True or False: The evolution of autotrophs was solely driven by a single catastrophic event affecting the environment.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the evidence supporting the transmission of impulses in plants, using the example of the sensitive plant, *Mimosa pudica*, to illustrate your argument.
12. Discuss the role of fermentation as an anaerobic process in cellular respiration, emphasizing how it regenerates NAD^+ and its significance when oxygen is scarce.

End of Paper